NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

# From 50 Steps to 1

## Distilling diffusion models for fast image generation

**La Nguyen Gia Hy • Le Duc Dong**
**Pham Viet Hoang • Nguyen Huu Thuat**

What must a student preserve when an iterative generator becomes a one- or few-step model?

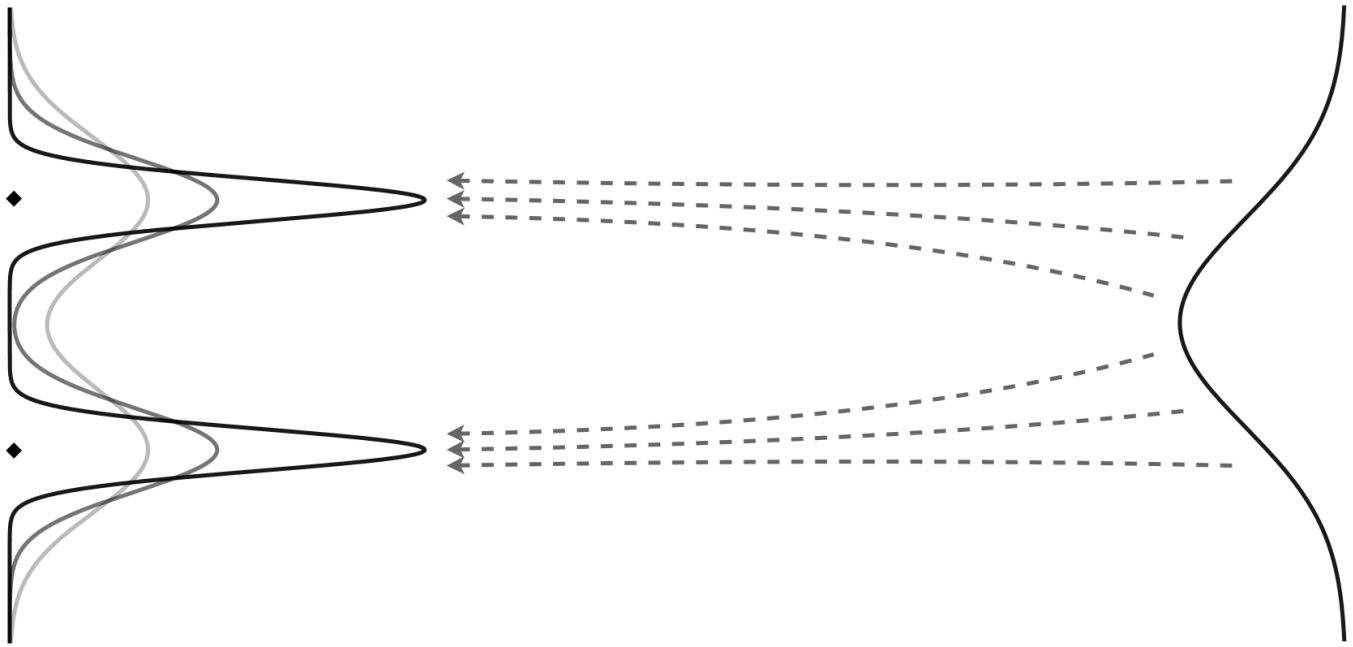
Consistency • Distribution matching • Adversarial refinement

Detailed methods, original author figures and controlled evidence | 2023–2026



SANA-Sprint • original author sample

# Distillation learns a different mapping



SDXL-Lightning: original probability-flow illustration

### Teacher

A pretrained vector field is integrated repeatedly.

### Student

Training changes the mapping so that much larger jumps become useful.

### NFE is not latency

Guidance branches, VAE decoding, batch size and hardware also affect cost.

COMMON NOTATION: CLEAN $x_0$, NOISE $xT$, TEXT $c$

$$x_t = \alpha_t x_0 + \sigma_t \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I)$$

ITERATIVE SAMPLING

$$\frac{dx_t}{dt} = v_\phi(x_t, t, c)$$

**One call to an unchanged denoiser is not the same as training a one-step generator.**

[1] Luo et al. · LCM · 2023 | [5] Lin et al. · SDXL-Lightning · 2024

# Related work: what supervision does the student receive?

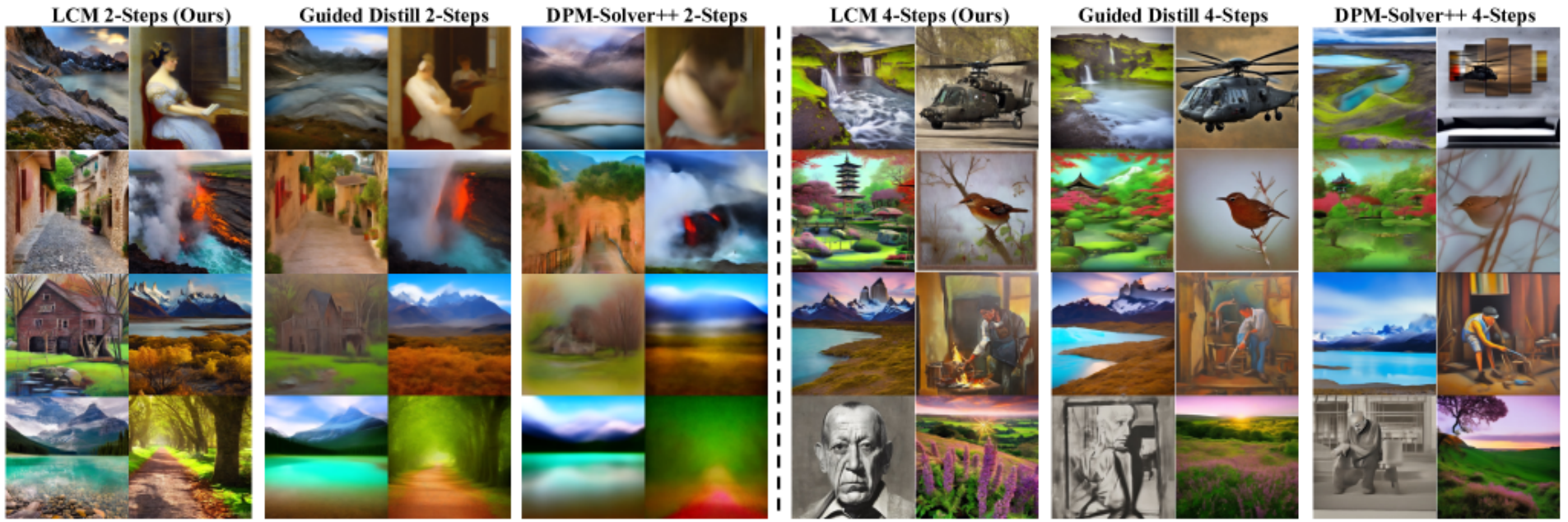
| What is matched? | Representative methods | Extra signal / central risk |
|---|---|---|
| Endpoint consistency | LCM (2023); SANA-Sprint (2025) | Teacher ODE targets; discretization and perceptual error |
| Generated distribution | DMD; SwiftBrush; SwiftBrush v2 (2024) | Real/fake scores; scorer tracking and mode coverage |
| Adversarial realism | SDXL-Lightning; DMD2 (2024) | Discriminator; data requirements and instability |
| Trajectory marginals / phases | TDM (2025); Phased DMD (2026) | Intermediate distributions; step-specific targets |
| Parameter or objective allocation | WaDi; Flash-DMD (2026) | Rotation adapters; timestep-specific losses + RL |

Synthesis of the cited methods, not mutually exclusive categories. LogCD (2026) adds local-to-global consistency.

**Compare the training signal—not just the publication year or sampling-step count.**

Sources: [1] [3] [6] [7] [8] [9]. Full primary references: slides 29–30.

# LCM learns a consistent endpoint in latent space




Original LCM qualitative comparison: two- and four-step outputs

## Guided teacher → stopped target

A teacher ODE solver supplies a lower-noise state. The student learns to agree with its EMA target.

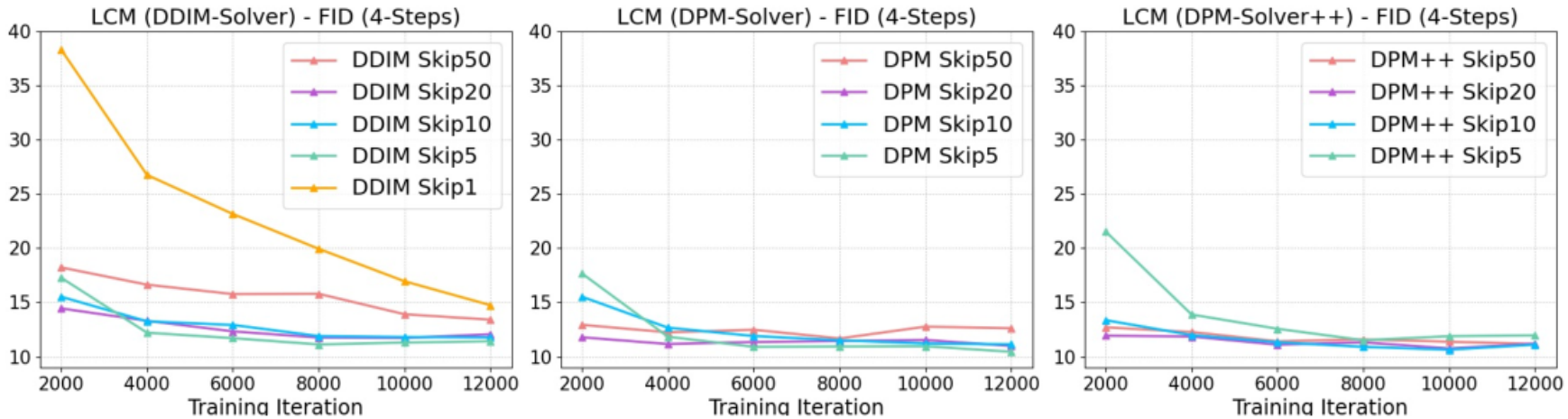
$$f_\theta(x_t, t, c) \approx f_\theta(x_s, s, c), \qquad t > s$$

$$\mathcal{L}_{\mathrm{CD}} = \mathbb{E}[\lambda_t \, d(f_\theta(x_t, t), \mathrm{sg}[f_{\bar{\theta}}(\hat{x}_s, s)])]$$

**Update the student; freeze the teacher. A shared endpoint predictor supports few-step refinement.**

[1] Luo et al. · LCM · 2023

# LCM: why four steps can beat one




Original solver / skipping-step ablation; every plotted model is evaluated at four steps

| Sampler | 1 step | 2 steps | 4 steps | 8 steps |
|---|---|---|---|---|
| DDIM | 183.29 | 81.05 | 22.38 | 13.83 |
| DPM++ | 185.78 | 72.81 | 18.43 | 12.20 |
| **LCM** | **35.36** | **13.31** | **11.10** | **11.84** |

Table 1, author-reported FID ↓. These LAION results must not be ranked against another paper's COCO FIDs.

### Read within this protocol

LAION-Aesthetic-6+, $512^2$, guidance $\omega$=8; LCM skipping k=20.

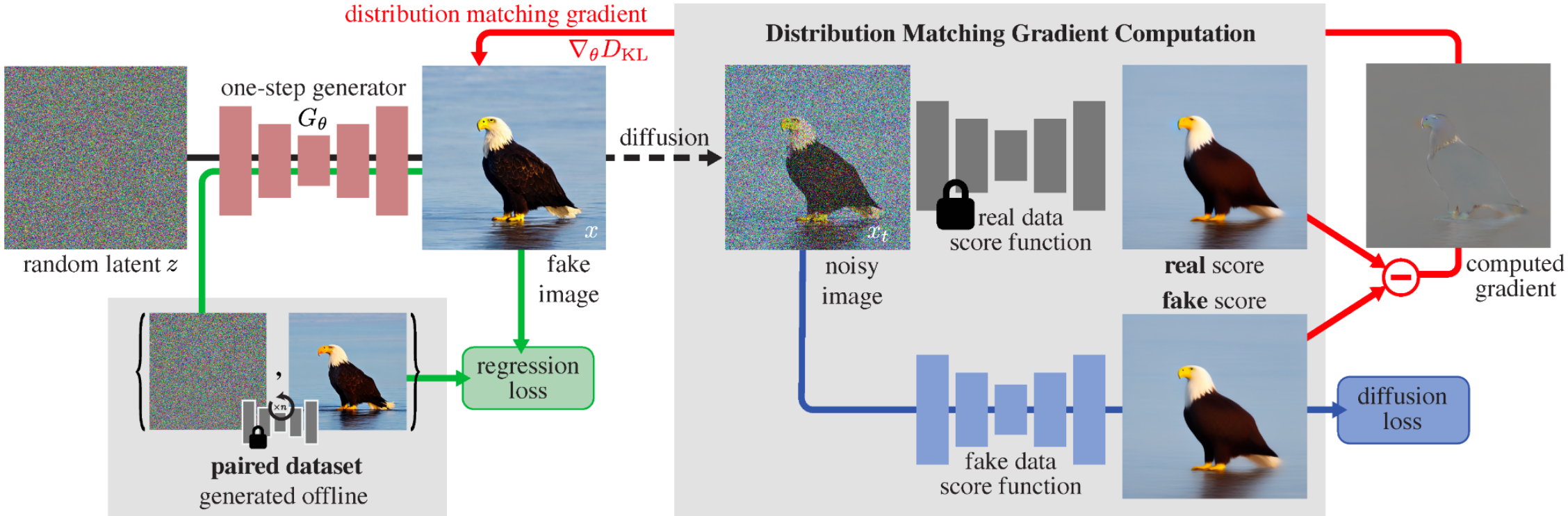
### Non-monotonic refinement

LCM FID: 11.10 at four steps versus 11.84 at eight.

**A distilled model's training target—not step count alone—determines its best operating point.**

[1] Luo et al. · LCM · 2023

# DMD: match distributions with two score models




Original DMD pipeline: generator, real/fake scores, and teacher-pair regression branch

## One-step generator

Noise becomes an image in one forward pass.

## Frozen real score

The pretrained diffusion model supplies the target-density direction.

## Moving fake score

An online scorer tracks the student's current distribution.

**Distribution matching need not reproduce the teacher's exact image for every noise seed.**

[2] Yin et al. · DMD · CVPR 2024

# DMD: the score difference supplies the update

NOISY-DISTRIBUTION TARGET

$$\mathcal{L}_{\mathrm{DM}} = \mathbb{E}_t \left[ w_t \, D_{\mathrm{KL}}(p_{\theta,t} \| p_{\phi,t}) \right]$$

HARMONIZED SCORE-FORM GRADIENT

$$\nabla_\theta \mathcal{L}_{\mathrm{DM}} = \mathbb{E}\left[ w_t \alpha_t (s_{\mathrm{fake}} - s_{\mathrm{real}}) \frac{\partial G_\theta}{\partial \theta} \right]$$

## Descent direction

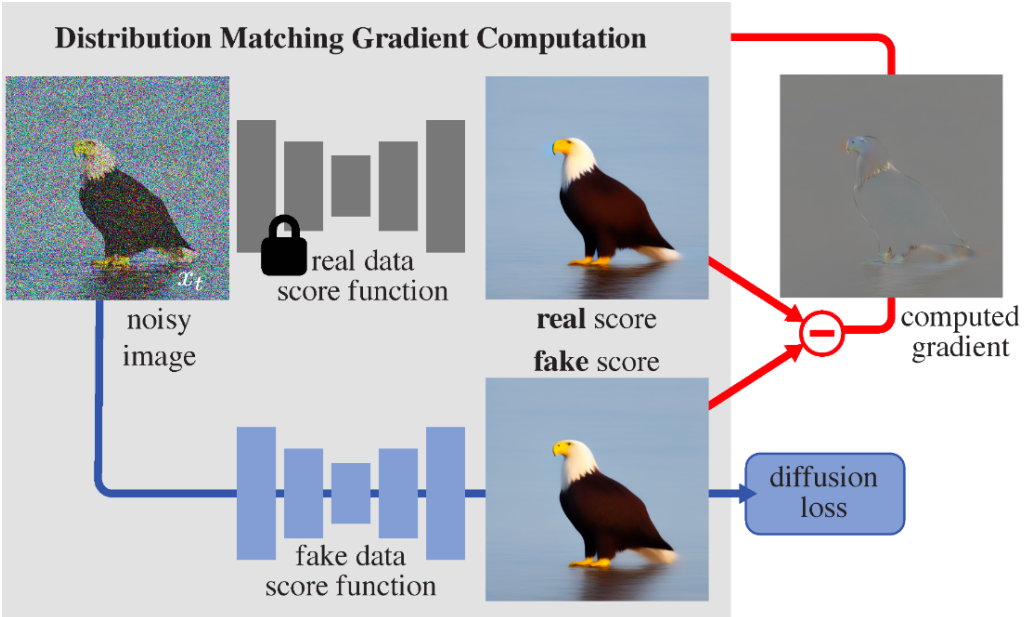
Gradient descent follows the real-minus-fake score direction.

## Regression stabilizer

LPIPS uses precomputed teacher noise–image pairs.

## Reported one-step result

SD 1.5 • 1 NFE • COCO-30K FID 11.49.




Crop of the original DMD scoring branch

SCORE MATCHING + PAIRED REGRESSION

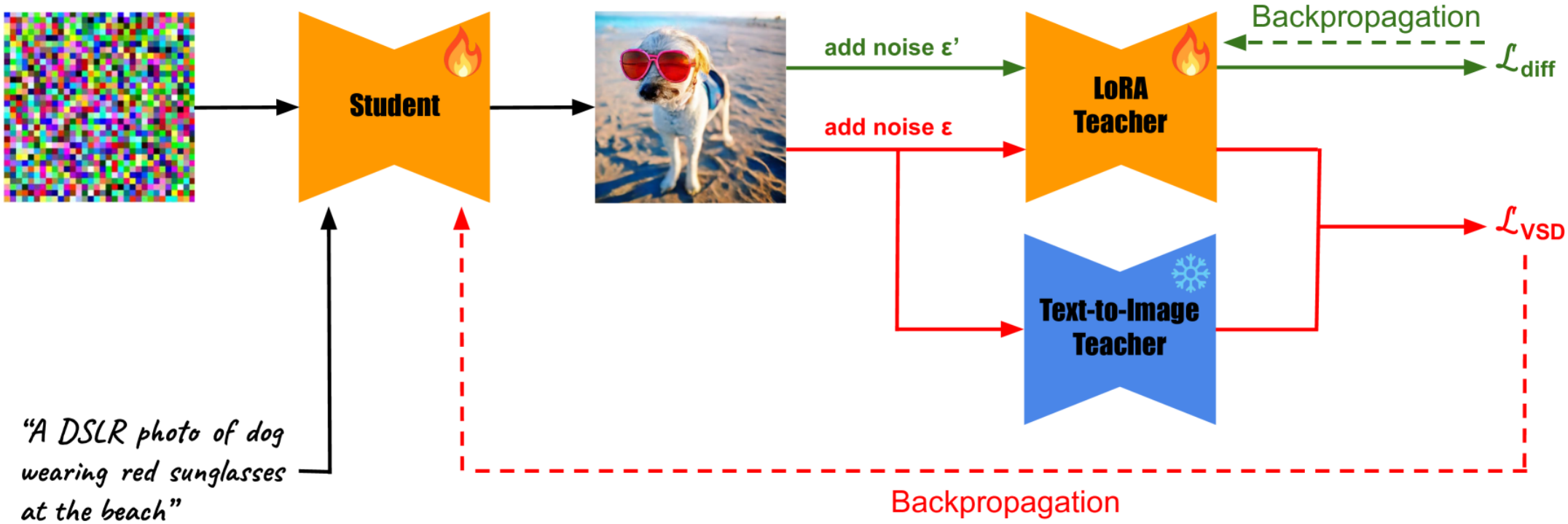
$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{DM}} + \lambda_{\mathrm{reg}} \mathcal{L}_{\mathrm{LPIPS}},$$

$$\mathcal{L}_{\mathrm{LPIPS}} = d\big(G_\theta(z), x_{\mathrm{teacher}}\big).$$

**Inference is one call. Training still uses two scorers and teacher-generated regression targets.**

[2] Yin et al. · DMD · CVPR 2024

# SwiftBrush: VSD without training images




Original SwiftBrush VSD pipeline; frozen teacher and trainable LoRA auxiliary teacher

STUDENT: GUIDED NOISE-PREDICTION DIFFERENCE

$$\nabla_\theta \mathcal{L}_{\text{VSD}} = \mathbb{E}\left[w_t(\hat{\epsilon}_\phi - \hat{\epsilon}_\psi)\frac{\partial G_\theta}{\partial \theta}\right]$$

AUXILIARY SCORER: DENOISING MSE

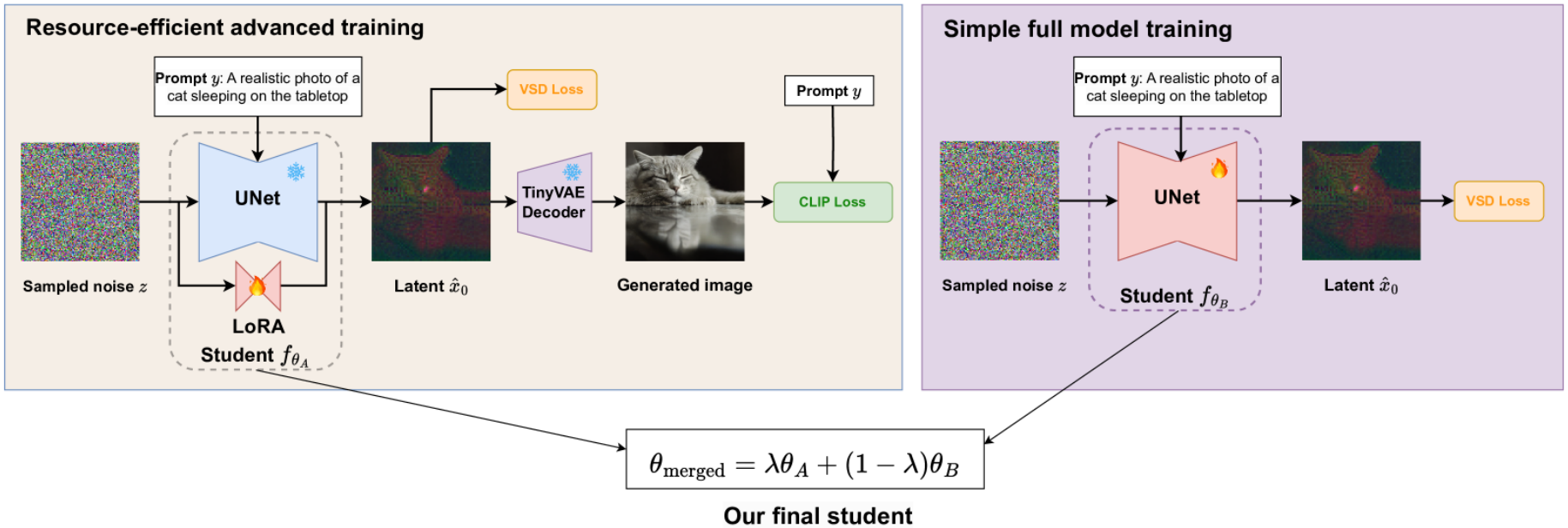
$$\min_\psi \mathbb{E}\left[\|\epsilon_\psi(x_t, t, c) - \epsilon\|_2^2\right]$$

Alternate scorer and student updates. Training uses text prompts and online synthetic samples—not cached teacher-image pairs.

**"Image-free" means no external training images; the procedure still generates samples during training.**

[3] Nguyen & Tran · SwiftBrush · CVPR 2024

# SwiftBrush v2: train, then merge two students




Original SwiftBrush v2 pipeline: parameter-efficient branch + full-model branch → weight fusion

## Initialization

Start from SD-Turbo, with SD 2.1 as the score teacher.

## Clamped CLIP signal

LoRA + TinyVAE uses a margin $\tau$=0.35; the CLIP weight decays to zero.

## One model at deployment

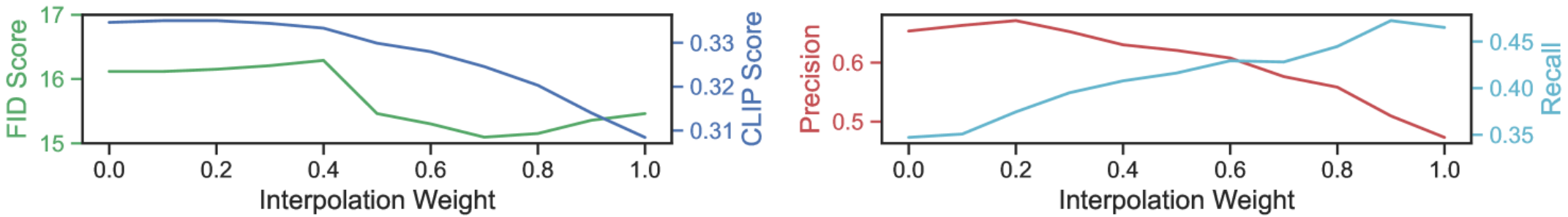
Fuse weights with $\lambda$=0.5. This is not a runtime ensemble.

$$\mathcal{L}_{\text{CLIP}} = \max\{0, \tau - \langle E_I(D(G_\theta)), E_T(c)\rangle\}$$

**The gain is a training-and-fusion recipe; deployment remains one step.**

[4] Dao et al. · SwiftBrush v2 · ECCV 2024

# SwiftBrush v2: quality is not the same as diversity




Original interpolation study: FID / CLIP and precision / recall respond differently to the weight

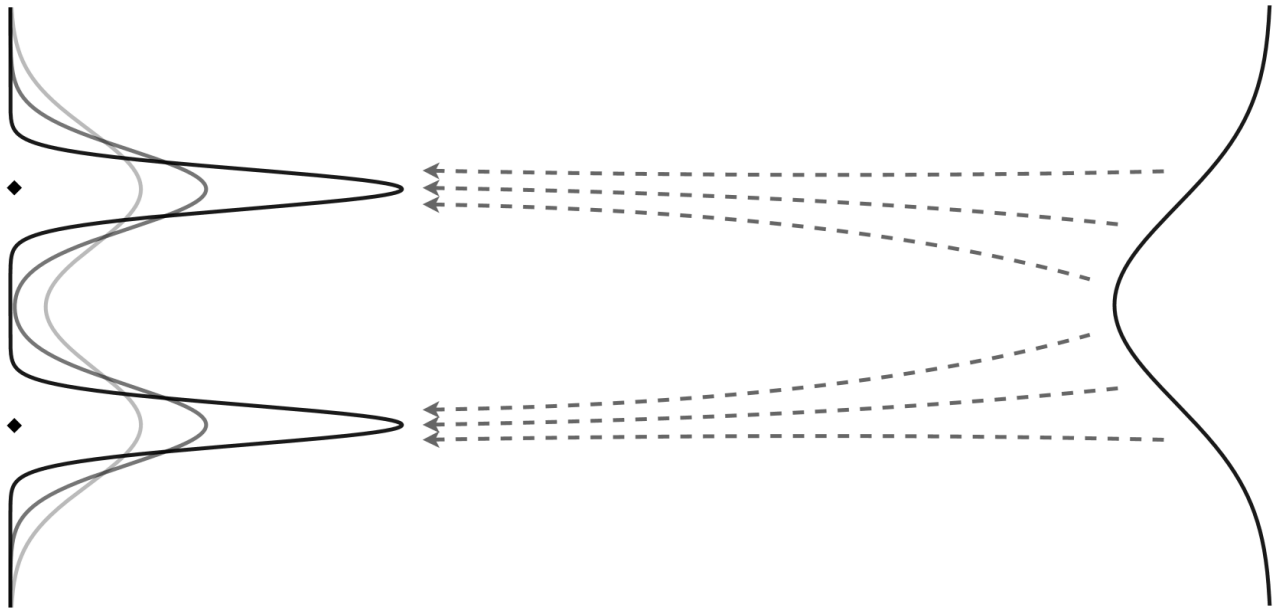
| Method | Steps | FID ↓ | Precision ↑ | Recall ↑ |
|---|---|---|---|---|
| SD 2.1, CFG 4.5 | 25 | 12.26 | 0.61 | 0.41 |
| SD-Turbo | 1 | 16.10 | 0.65 | 0.35 |
| SwiftBrush (v2 rerun) | 1 | 15.46 | 0.47 | 0.46 |
| **SwiftBrush v2, image-free** | **1** | **8.77** | **0.55** | **0.53** |
| **SwiftBrush v2*, + image regularization** | **1** | **8.14** | **0.57** | **0.52** |

Same v2 evaluation: COCO-2014, first 30K prompts, FID on resized $256^2$ outputs. * adds 200K image pairs; it is not the image-free result.

**8.77 is the image-free FID. The 8.14 result includes additional image regularization.**

[4] Dao et al. · SwiftBrush v2 · ECCV 2024

# SDXL-Lightning: progressive adversarial distillation



Original flow-capacity illustration; not an invented training pipeline

PROGRESSIVE STEP-COUNT CURRICULUM

$$128 \longrightarrow 32 \longrightarrow 8 \longrightarrow 4 \longrightarrow 2 \longrightarrow 1$$

## Progressive curriculum

128→32 uses MSE; subsequent stages use adversarial distillation.

## Conditional discriminator

Initially compares transitions while retaining their starting-state correspondence.

## Unconditional refinement

Removing the starting-state condition relaxes rigid flow matching and reduces artifacts.

GENERIC ADVERSARIAL GENERATOR OBJECTIVE

$$\mathcal{L}_G = -\mathbb{E}[\log D(G_\theta(\cdot))]$$

**Adversarial supervision relaxes exact path correspondence while preserving perceptual realism.**

[5] Lin et al. · SDXL-Lightning · 2024

# Lightning: patch quality and global FID disagree

| Method | Steps | Whole-image FID ↓ | Patch FID ↓ | CLIP ↑ |
|---|---|---|---|---|
| SDXL teacher | 32 | 18.49 | 35.89 | 26.48 |
| SDXL-Lightning | 1 | 22.61 | 41.53 | 26.02 |
| SDXL-Lightning | 2 | 23.11 | 35.12 | 25.98 |
| **SDXL-Lightning** | **4** | **22.30** | **33.52** | **26.07** |
| SDXL-Lightning | 8 | 21.43 | 33.55 | 25.86 |

Paper Table 2 · COCO first 10K validation prompts. Whole image resized to 299px; patch FID uses a 299px center crop. All numbers are author-reported.

## Four-step evidence

Patch FID improves over the 32-step teacher; whole-image FID does not.

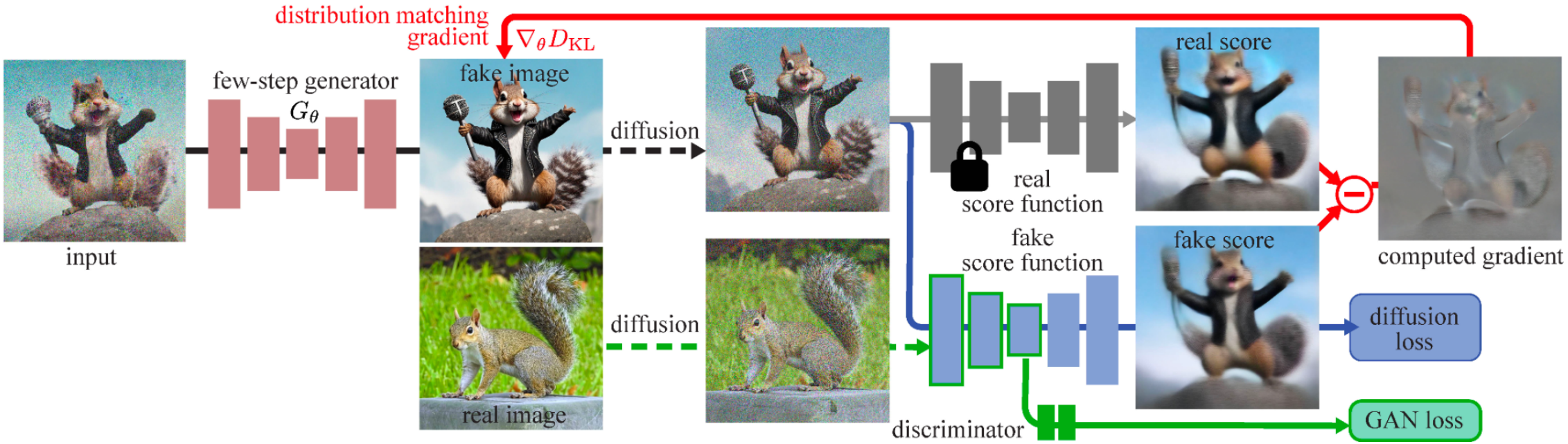
## Deployment constraint

Use the intended checkpoint step count and prediction type. The one-step release is experimental.

**“Better than the teacher” must name the quality dimension and evaluation protocol.**

[5] Lin et al. · SDXL-Lightning · 2024

# DMD2: remove regression, strengthen the training system




Original DMD2 pipeline: distribution matching + denoising loss + real-data GAN

## No cached teacher pairs

The LPIPS regression branch from original DMD is removed.

## Faster scorer tracking

Five fake-score updates per generator update improve critic tracking.

## GAN + inference states

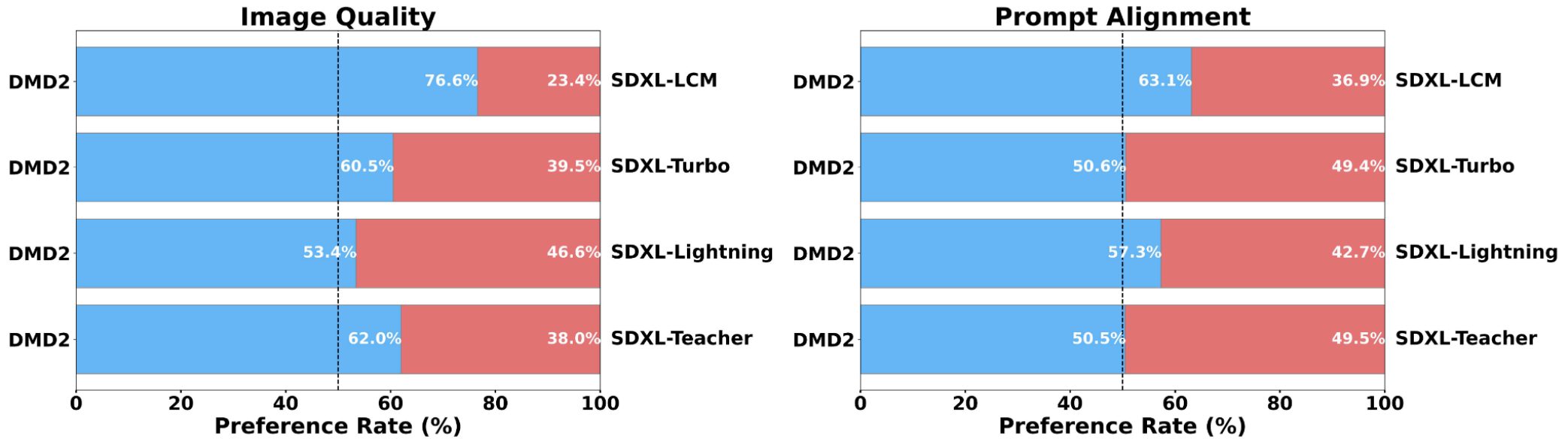
Use real-image supervision and multistep training simulation.

$$\mathcal{L}_G = \mathcal{L}_{\mathrm{DM}} + \lambda_{\mathrm{adv}}\mathcal{L}_{\mathrm{GAN}}$$

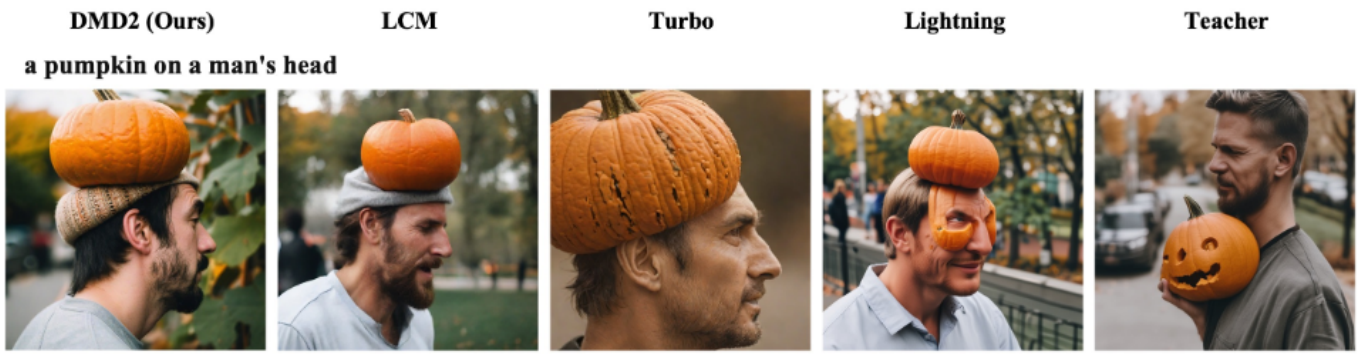
**Removing teacher-image pairs does not make the full DMD2 method image-free.**

[6] Yin et al. · DMD2 · NeurIPS 2024

# DMD2: preferences and sample quality




Original SDXL human-preference study: image quality and prompt alignment are distinct judgments




Original author-selected prompt row, cropped with its model headings. The complete comparison image is in the source package.

## Preserve the labels

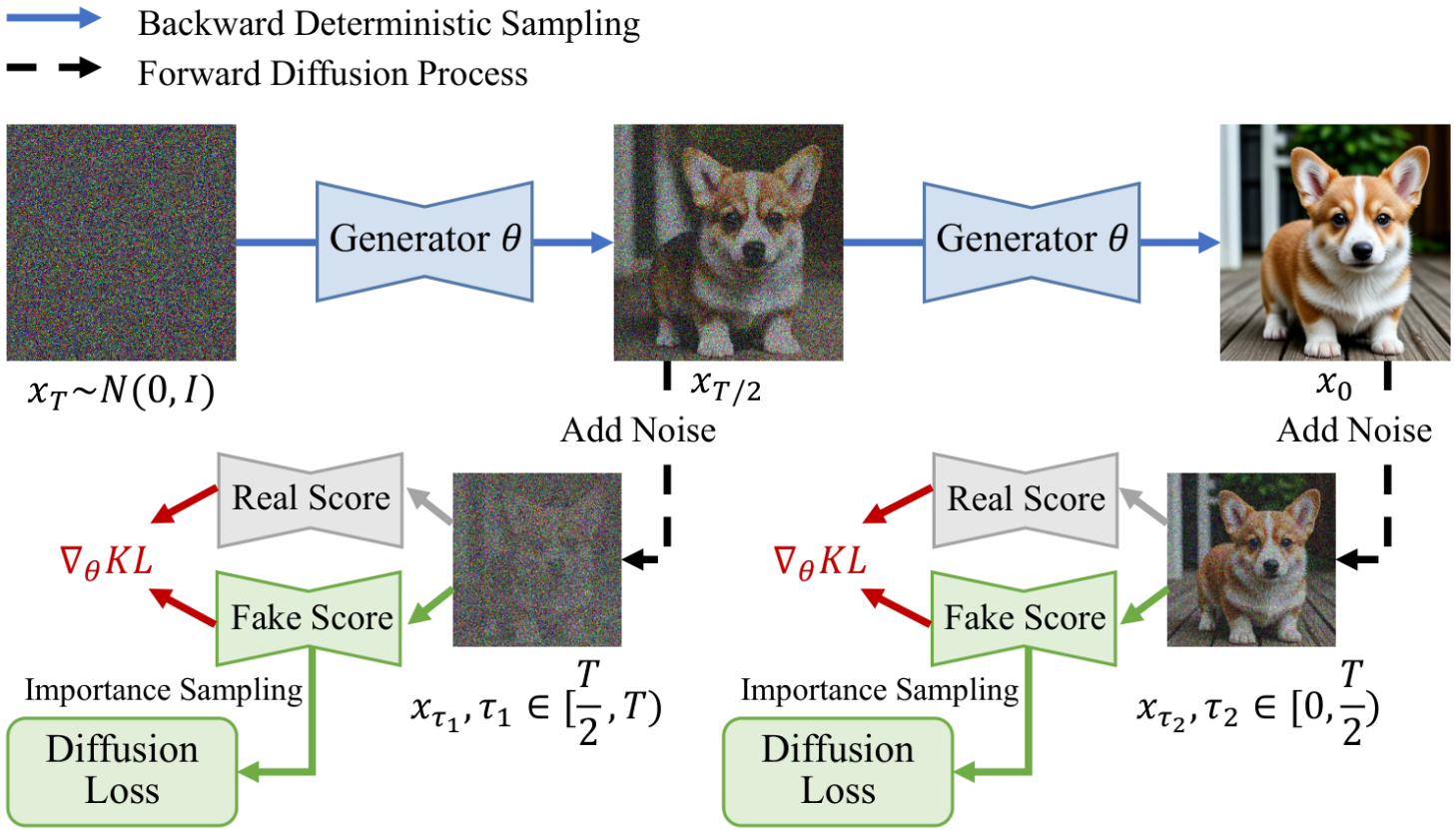
The human study is an SDXL experiment.

## Do not mix result tracks

FID 8.35 in the source refers to SD 1.5, not this SDXL study.

[6] Yin et al. · DMD2 · NeurIPS 2024

# TDM supervises distributions along the trajectory




Original TDM pipeline; intermediate states receive distribution supervision

## Intermediate distributions

Match teacher and student marginals, not identical sample paths.

## Step-aware fake score

A pooled scorer otherwise estimates a mixture of different steps.

## Decoupled training

Sample a trajectory segment, fit its score, then update the generator.

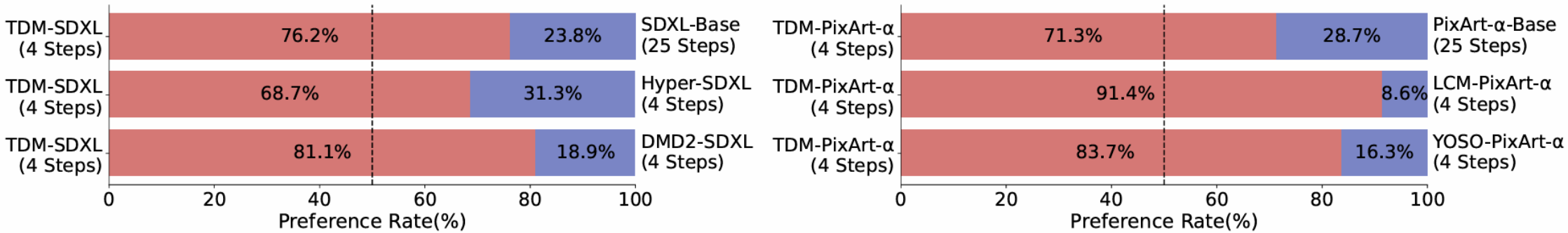
WHY A POOLED SCORER IS NOT STEP-SPECIFIC

$$s_{\text{mix}} = \frac{p_1 s_1 + p_2 s_2}{p_1 + p_2}$$

**Trajectory information can supervise distribution matching without exact path regression.**

[7] Luo et al. · TDM · ICCV 2025

# TDM: evaluate image preference and video separately




Original user-preference figure; teacher labels show 25 sampling steps, not 50 guided NFEs

| Method | Total Score↑ | Quality Score↑ | Semantic Score↑ |
|---|---|---|---|
| CogVideoX-2B | 80.91 | 82.18 | 75.83 |
| TDM (4 NFE) | **81.65** | **82.66** | **77.64** |

Original video benchmark table • CogVideoX-2B / TDM

The reported distillation budget excludes teacher pretraining. A preference win does not establish universal visual superiority.

## Image distillation budget

PixArt-$\alpha$: 50 teacher NFE $\rightarrow$ 4 student NFE. The source reports 500 iterations and 2 A800 GPU-hours.

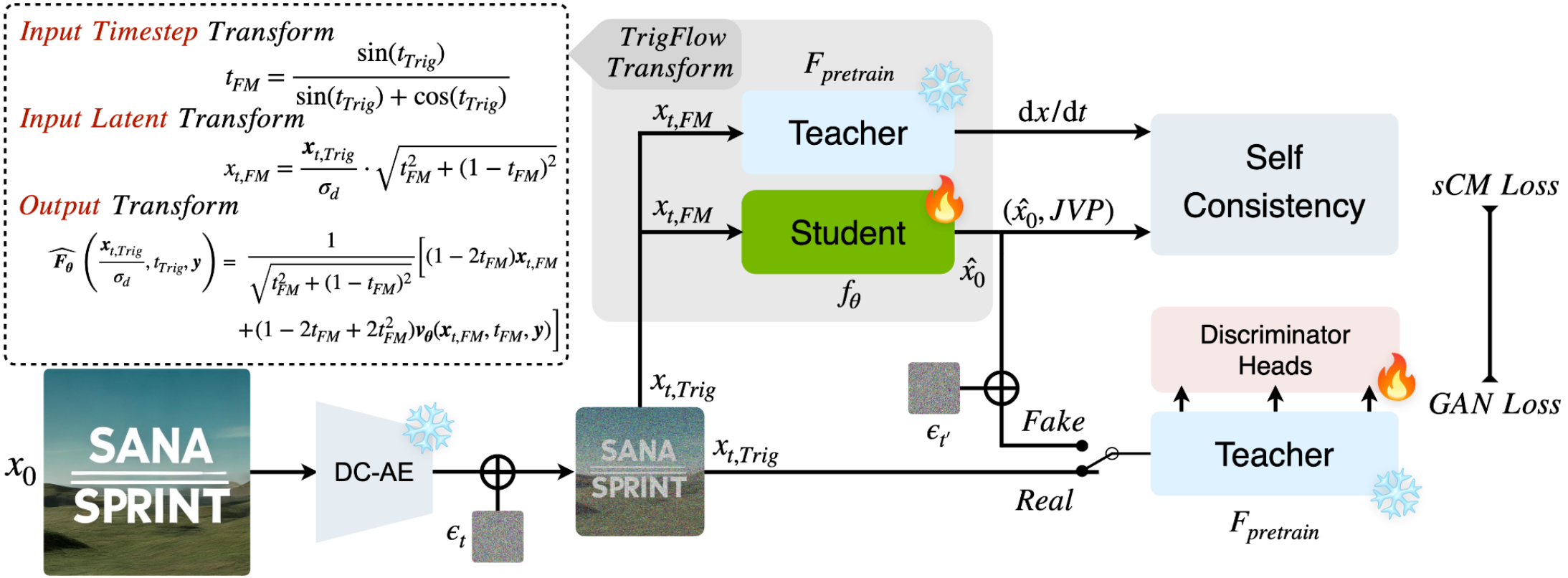
## Separate video experiment

CogVideoX-2B: 100 $\rightarrow$ 4 NFE; VBench 80.91 $\rightarrow$ 81.65. Do not transfer the image training budget to video.

**Attach the backbone, task, sampling convention and training budget to every headline number.**

[7] Luo et al. · TDM · ICCV 2025

# SANA-Sprint: align schedules, then combine losses




Original SANA-Sprint pipeline: TrigFlow transfer, continuous-time consistency and latent adversarial supervision

TRIGFLOW PARAMETERIZATION

$$x_t = \cos(t)x_0 + \sin(t)\epsilon, \qquad t \in [0, \pi/2]$$

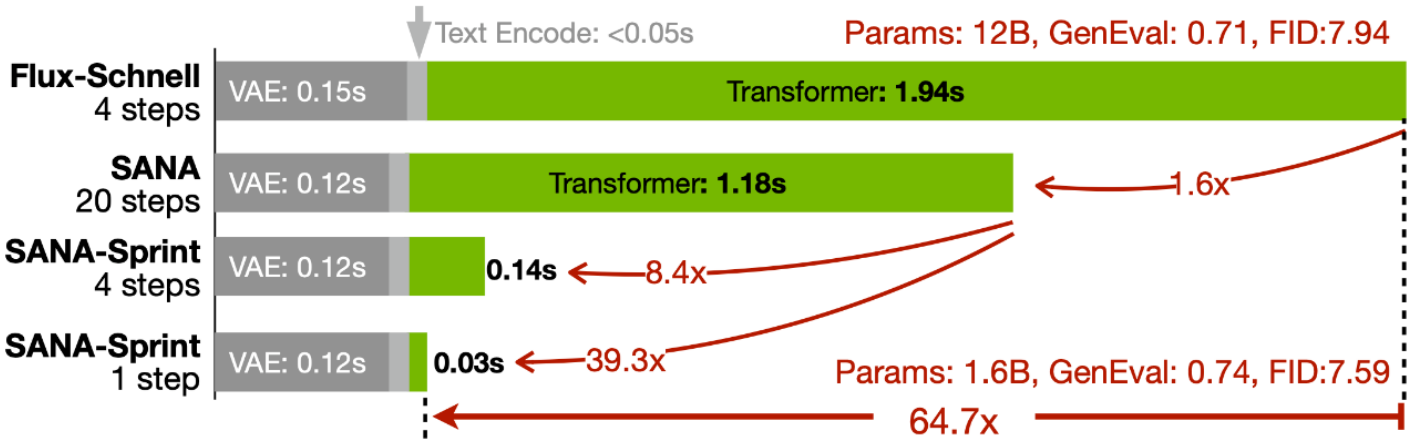
OBJECTIVE-FAMILY SUMMARY

$$\mathcal{L} = \mathcal{L}_{\text{sCM}} + \lambda\mathcal{L}_{\text{LADD}}$$

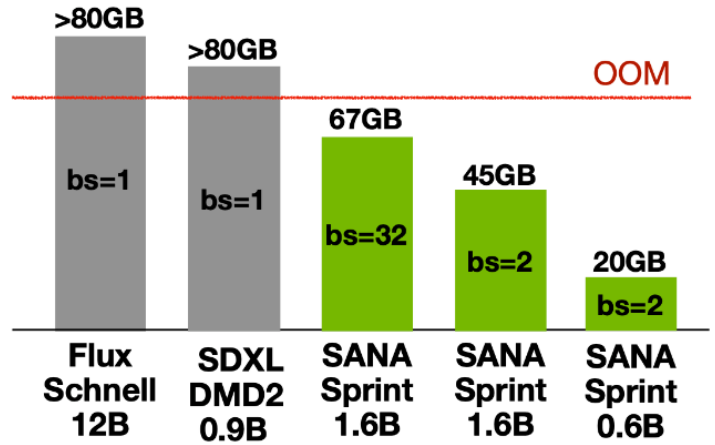
**Schedule compatibility comes first; consistency and adversarial realism then supply complementary signals.**

[8] Chen et al. · SANA-Sprint · ICCV 2025

# SANA-Sprint: separate the ablation from the speed claim




(a) Generation latency (s) on 1024x1024




(b) Training GPU Memory (GB)

Original latency / training-memory figure. Memory bars use different batch sizes, as labeled.

| Training objective | FID ↓ | CLIP ↑ |
|---|---|---|
| sCM only | 8.93 | 27.51 |
| LADD only | 12.20 | 27.00 |
| **sCM + LADD** | **8.11** | **28.02** |

Table 3 • two-step analysis: start $\pi/2$, intermediate t=1.0.

## Controlled ablation

The table is the two-step loss study, not the one-step headline result.

## Read the time components

The figure's 0.03 s is transformer time; VAE and text encoding are separate.

**Fewer NFEs help, but architecture, decoding and batch size determine real system cost.**

[8] Chen et al. · SANA-Sprint · ICCV 2025

# Before 2026: inference is cheap; supervision is not free

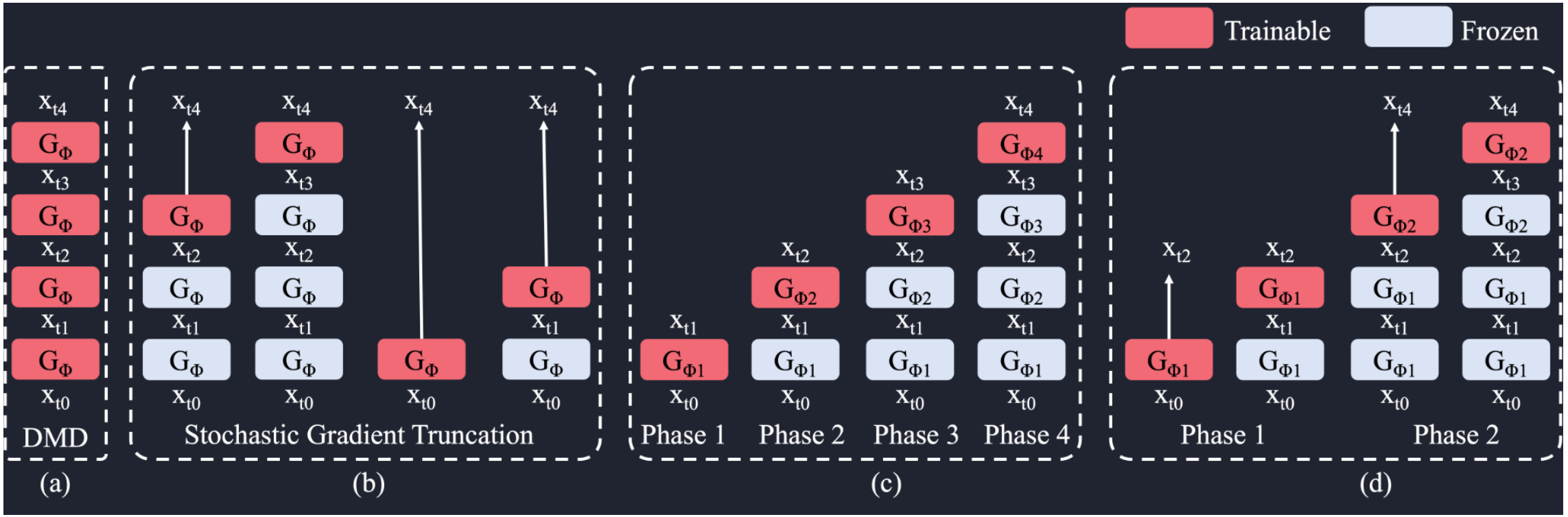
| Method | External signal during distillation | Deployment implication |
|---|---|---|
| LCM | Training image latents + guided ODE targets | Consistency model; few-step refinement |
| DMD | Teacher-generated regression pairs + two scores | One generator evaluation |
| SwiftBrush / v2 base | Prompts + online synthetic samples; v2 merges weights | One-step student; no runtime ensemble |
| Lightning | Training data + progressive teacher/discriminator | Step-specific checkpoints |
| DMD2 | Real-image GAN + online score fitting | One-/few-step student; heavier critic training |
| TDM | Student trajectory + teacher / fake score models | Few-step trajectory; no cached teacher-image pairs |
| SANA-Sprint | Pretrained flow + consistency / adversarial signals | Unified 1–4-step model |

Training-signal summary, not a common-dataset ranking. SwiftBrush v2* additionally uses image regularization.

**Separate distillation-time data and modules from what is needed at inference.**

Sources: [1] [2] [3] [4] [5] [6] [7] [8]. Full primary references: slides 29–30.

# Phased DMD: specialize across noise intervals




Original four-panel comparison: full unrolling, gradient truncation, per-step phases and grouped phases

## Partition the learning problem

Match progressively cleaner intermediate distributions over SNR subintervals.

$$\min_{\theta} \; D_{\mathrm{KL}}(p_{\theta,t_i} \| p_{\phi,t_i})$$

## Preserve phase specialization

Freeze earlier learned structure while a later expert models detail.

**A phase is a distinct learning problem—not simply another Euler update.**

[9] Fan et al. · Phased DMD · CVPR 2026

# Phased DMD: diversity gains depend on the metric

| Backbone / metric | Teacher | DMD2 | Phased DMD |
|---|---|---|---|
| Wan2.2 image diversity: DINOv3 ↓ | 0.732 | 0.828 | 0.768 |
| Wan2.2 image diversity: LPIPS ↑ | 0.531 | 0.447 | 0.481 |
| Qwen-Image diversity: DINOv3 ↓ | 0.907 | 0.941 | 0.958 |
| Qwen-Image diversity: LPIPS ↑ | 0.483 | 0.309 | 0.322 |
| Wan2.2 T2V motion: optical flow ↑ | 10.26 | 3.23 | 9.30 |

Image diversity: 21 prompts × 8 seeds; teacher 40 steps/CFG4, students 4 steps/CFG1. Video motion: separate 220-prompt, fixed-seed evaluation.

### Wan2.2

Image diversity and motion measures move closer to the teacher in these experiments.

### Qwen-Image counterexample

LPIPS improves over DMD2, but DINOv3 similarity rises—worse diversity under that metric.

**Do not hide a disagreement between metrics behind a single “improves diversity” claim.**

[9] Fan et al. · Phased DMD · CVPR 2026

# LogCD: local-to-global consistency

## First, learn local consistency

Local consistency is learned before enforcing agreement along the complete inference path.

## Then, improve perceptual consistency

The cited abstract describes a latent learned perceptual image-patch similarity model.

## One model, several step counts

The related-work source reports a unified model for two to four sampling steps.

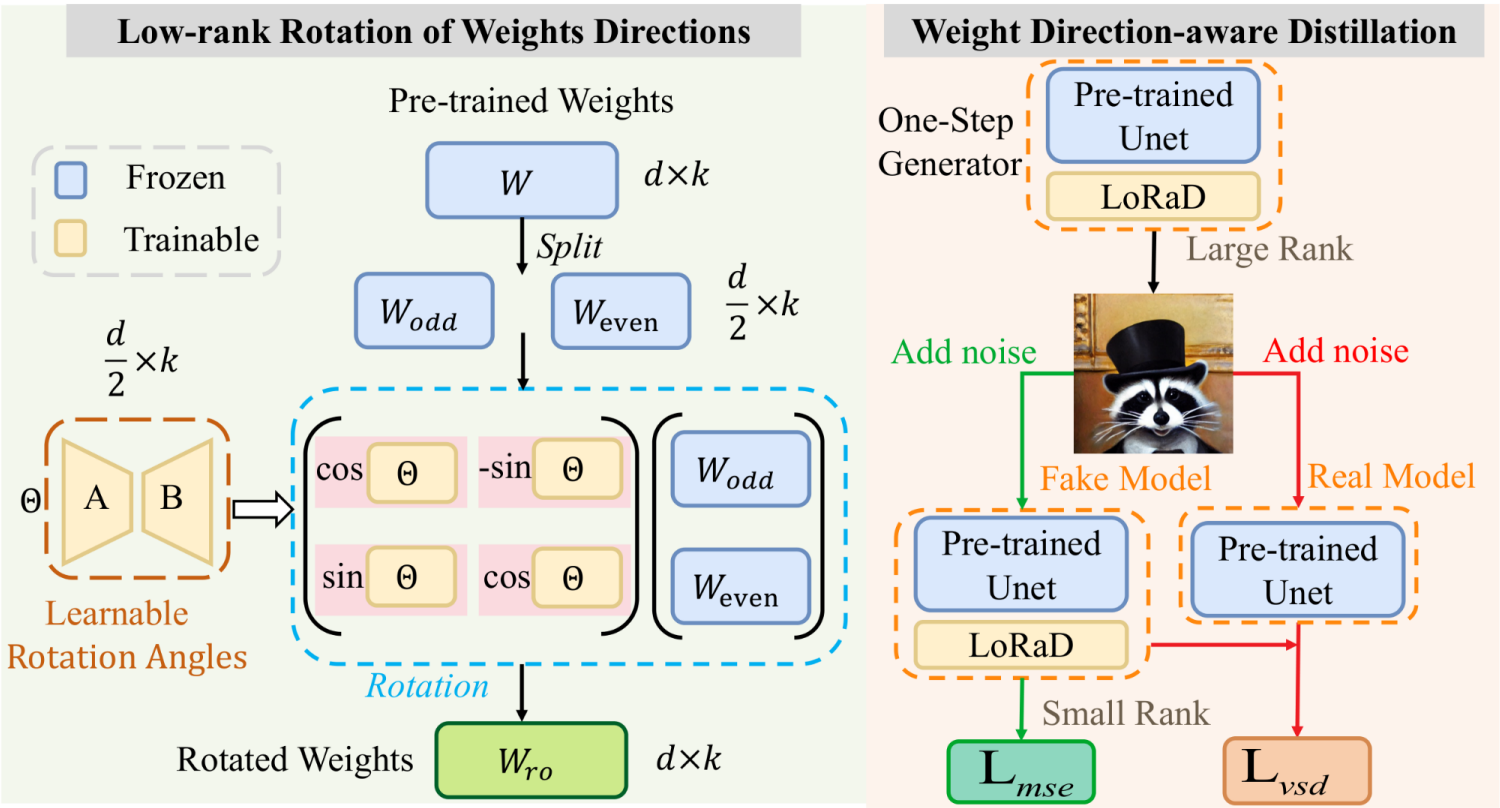
| Reported in the cited abstract | Value |
|---|---|
| SDXL sampling steps | 3 |
| CLIP score | 33.5 |
| Distillation budget | 70 A100 GPU-hours |

This entry remains abstract-level. No original pipeline or exact loss was supplied, so neither is invented here.

**Local supervision and global trajectory consistency are complementary—not interchangeable.**

[10] Xie et al. · LogCD · CVPR 2026

# WaDi: rotate pretrained weight directions




Original WaDi pipeline: LoRaD rotation (left), VSD framework (right)

## Direction matters

The source's replacement study finds direction changes more consequential than norm changes.

## Low-rank angle matrix

LoRaD parameterizes pairwise rotation angles, rather than an additive weight update.

## VSD with adapters

Train rotation adapters for student and fake score; keep the teacher fixed.

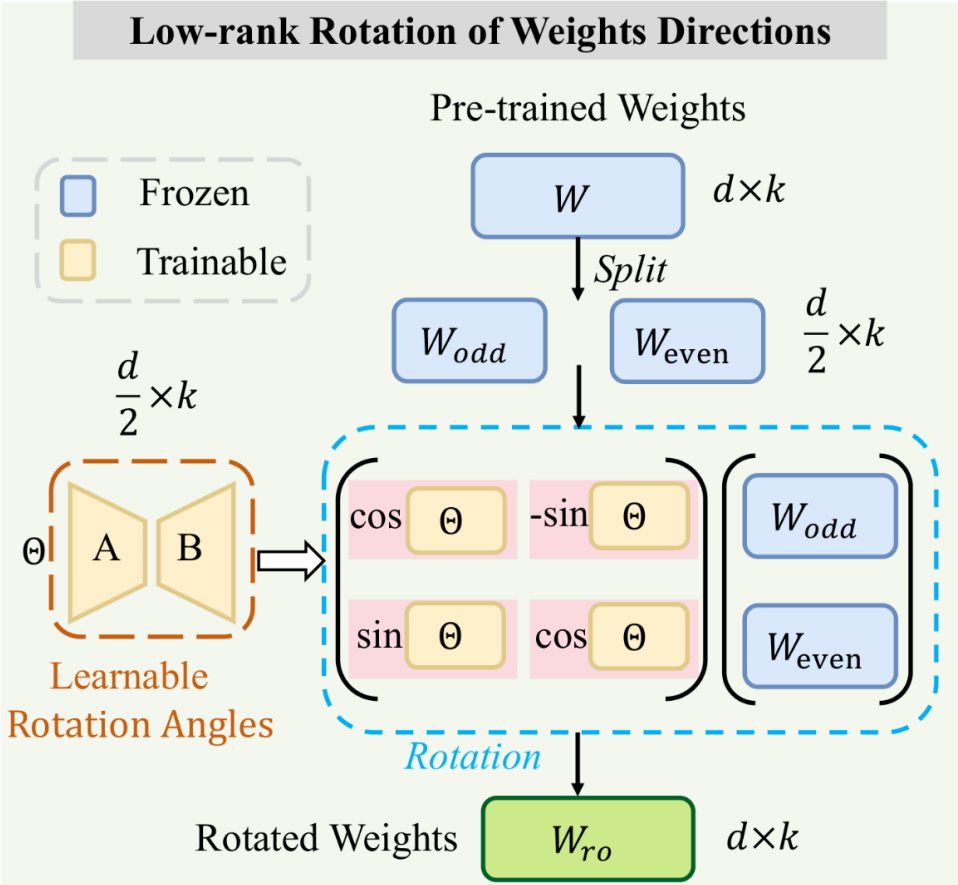
PAIRWISE ROTATION SUMMARY

$$\Theta = AB, \quad R(\vartheta)^\top R(\vartheta) = I$$

**Orthogonal pairwise rotations preserve norms while changing the directions the student can adapt.**

[11] Wang et al. · WaDi · CVPR 2026

# WaDi: more adapter rank is not always better




Crop of the original LoRaD panel

| Student rank | Student parameters | Fake rank | FID ↓ |
|---|---|---|---|
| 64 | 20.95M | 32 | 13.64 |
| 128 | 41.90M | 32 | 13.16 |
| **256** | **83.80M** | **32** | **10.79** |
| 512 | 167.59M | 32 | 12.75 |

WaDi Table 3 · SD 1.5 / VSD / COCO-2014. Fake model at rank 32 has 9.38M parameters. These are trainable counts, not total model sizes.

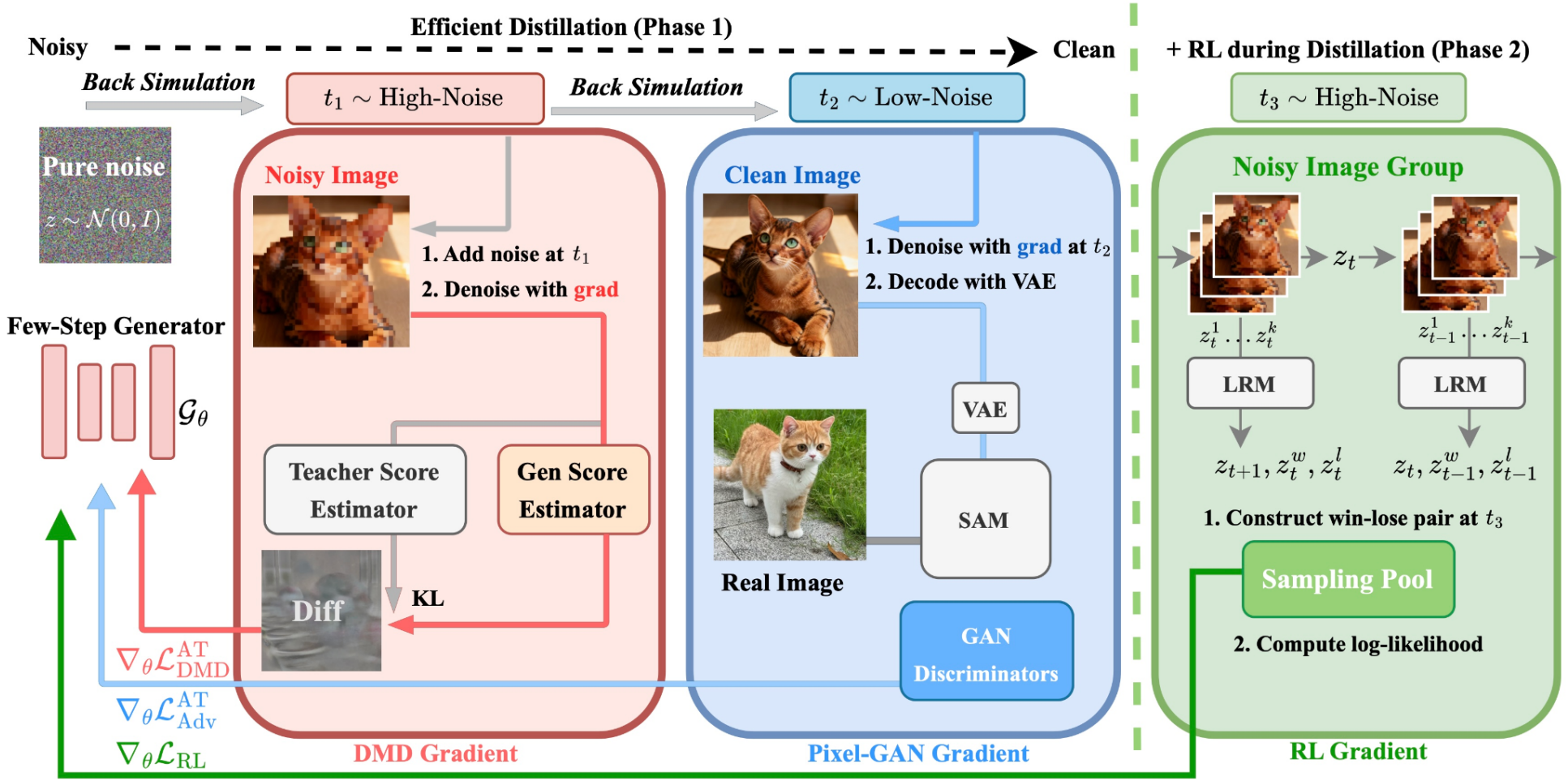
## Parameter count is not total compute

83.80M counts trainable student parameters, not the frozen backbone or the complete training system.

**Rank 256 is best in this sweep; doubling it to 512 worsens the reported FID.**

[11] Wang et al. · WaDi · CVPR 2026

# Flash-DMD: noise level determines the training signal




Original Flash-DMD pipeline: high-noise DMD, low-noise pixel GAN, then joint RL during distillation

**Distillation remains active during preference optimization rather than ending before RL begins.**

[12] Chen et al. · Flash-DMD · CVPR 2026

# Flash-DMD: read the denominator behind 2.1%

| SDXL / 4 NFE | ImageReward ↑ | CLIP ↑ | PickScore ↑ | Reported budget |
|---|---|---|---|---|
| DMD2 baseline | 0.8748 | 0.3302 | 0.2309 | 128 × 24K |
| **Flash, stage 1 / TTUR1-1K** | **0.9509** | **0.3292** | **0.2322** | **64 × 1K** |
| Flash, stage 1 / TTUR2-8K | 0.9740 | 0.3298 | 0.2327 | 64 × 8K |

Paper Table 1 · COCO-10K. Cost column uses its reported training-budget units; 2.1% is the 64K/3072K ratio, not a universal wall-clock or energy ratio.

REPORTED STAGE-1 TRAINING-BUDGET RATIO

$$\frac{64 \times 1{,}000}{128 \times 24{,}000} = 2.083\%$$

GENERATOR-TO-SCORER EMA: SCHEMATIC UPDATE

$$\psi \leftarrow \lambda_{\text{ema}}\psi + (1 - \lambda_{\text{ema}})\theta$$

## What the comparison supports

ImageReward and PickScore improve; CLIP does not improve in these rows.

## What the ratio does not establish

This is not a universal GPU-hour, wall-clock or energy ratio. Later RL adds training cost.

**A cost percentage is meaningful only with its baseline, accounting units and training stage.**

[12] Chen et al. · Flash-DMD · CVPR 2026

# Inference recap: one step vs. a few steps

**FEW-STEP / FLEXIBLE**

## 1–4 steps

Allow a few refinement opportunities instead of forcing one giant jump.

LCM · TDM · SANA-Sprint · LogCD · Lightning

**ONE-STEP FOCUSED**

## 1 step

Minimum model calls, but the strongest compression constraint.

DMD · SwiftBrush · SwiftBrush v2 · WaDi

### SANA-Sprint — quality across 1–4 steps

0.6B model · same reported setting · FID ↓ / GenEval ↑

| Metric | 1 step | 2 steps | 4 steps |
|---|---|---|---|
| FID ↓ | **7.04** | **6.54** | **6.48** |
| GenEval ↑ | **0.72** | **0.76** | **0.76** |

### DMD2 — whole-image vs. patch quality

SDXL · COCO-2014 10K · same protocol · lower is better

| Metric | 1 step | 4 steps |
|---|---|---|
| FID ↓ | **19.01** | **19.32** |
| Patch FID ↓ | **26.98** | **20.86** |

**LCM — more steps are not always better**

| | 1 step | 2 steps | 4 steps | 8 steps | |
|---|---|---|---|---|---|
| FID ↓ | **35.36** | **13.31** | **11.10** | **11.84** | **Best at 4 steps** |

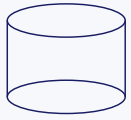
**One step can already be strong; a few extra steps can still help — but more steps are not automatically better.**

Sources: [8] SANA-Sprint; [6] DMD2 Table 2; [1] LCM Table 1. Protocols differ across examples; compare within each example.

# Different methods save different training resources

DATA / PAIRS

## SwiftBrush



**No external training images**

No cached teacher-image pairs

Uses online synthetic samples instead.

**Saves** **data and pair-preparation requirements**

DISTILLATION COST

## TDM

≈ **2** **A800 GPU-hours**

PixArt-$\alpha$

**50 teacher steps** → **4 student steps**

Reported 500 iterations.

**Saves** **low reported distillation cost**

TRAINABLE PARAMETERS

## WaDi

**860M** total U-Net params



**≈ 84M trained** **~10%**

Still targets a strong one-step student.

**Saves** **the number of parameters updated**

**Overall: preserve the capabilities we need at an inference and training budget we can actually afford.**

Sources: [3] SwiftBrush; [7] TDM; [11] WaDi. Full primary references: slides 29–30.

# Primary sources | 2023–2024


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